

Ethan Chau

echau18@cs.uw.edu

Web: echau18.gitlab.io | LinkedIn: echau18 | GitHub: ethch18

Experience

Microsoft | Senior Applied Scientist (Search Relevance AI/ML)

Jan. 2022 – Present | Redmond, WA

Senior Applied Scientist (Sept. 2023 – Present)

Applied Scientist 2 (Sept. 2022 – Aug. 2023)

Applied Scientist (Jan. 2022 – Aug. 2022)

- Building Bing's core **dense retrieval** model that encodes queries into vectors for web search, delivering consistent **+2-6% weighted recall** improvements in every release through state-of-the-art **NLP/deep learning** techniques.
- Shipped Bing's first **multi-vector** retriever with **+6% weighted recall** over single-vector models and **96% less** storage than naïve multi-vector models.
- Developing the critical integrations between science and platform teams that maintains data freshness.
- Co-mentored four interns with MSR in experimenting and publishing [1][2][3].
- Coordinated across platform, hardware, and science teams to lead an organization-wide infrastructure migration that increased **training capacity 2x**.
- Greatly exceeded expectations in every annual performance review.

University of Washington | NLP Researcher

Mar. 2018 – Nov. 2021 | Seattle, WA | Advisor: Prof. Noah A. Smith

- Led the design of informed training methods for **low-resource** contextual word **representation**, with up to **39%** error reduction on Latin-script parsing evaluations [5] and a **57%** reduction on non-Latin part-of-speech tagging [4].

University of Washington | Teaching Assistant (ML, NLP)

Jan. 2019 – Mar. 2019 and Sept. 2020 – June 2021 | Seattle, WA

- Developed a new **PyTorch**- and AllenNLP-based text classification assignment.
- Published a comprehensive, officially featured tutorial for AllenNLP. ([link](#))

Facebook | Software Engineering Intern (Machine Learning)

June 2020 – Sept. 2020 | Seattle, WA

- Drove the development of a configurable, E2E system for content **retrieval** and recommendation in **C++** and **Hack**, increasing engagement by **2.5%**.
- Designed efficient data-collection pipelines for **search index** construction.
- Contributed to the **PyTorch** open-source project.
- Achieved highest possible rating ("**Rockstar**") on final performance evaluation.

Facebook | Software Engineering Intern (Backend)

June 2019 – Sept. 2019 | Menlo Park, CA

- Architected a modular, interpretable, and configurable suggestion system in **Hack** with a **37%** higher success rate than previous versions.

Selected Publications

- [1] G. Qin, C. Rosset, **E. C. Chau**, N. Rao, and B. Van Durme. 2024. Dodo: Dynamic Contextual Compression for Decoder-only LMs. In *Proc. Of ACL*. ([link](#))
- [2] C. Xu, C. Rosset, **E. C. Chau**, et al. 2024. Automatic Pair Construction for Contrastive Post-training. In *Findings of ACL: NAACL*. ([link](#))
- [3] C. Rosset, H.-L. Chung, G. Qin, **E. C. Chau**, et al. 2025. Researchy Questions: A Dataset of Multi-Perspective, Decompositional Questions for LLM Web Agents. In *SIGIR*. ([link](#))
- [4] **E. C. Chau** and N. A. Smith. 2021. Specializing Multilingual Language Models: An Empirical Study. In *Workshop on Multilingual Representation Learning (EMNLP)*. (**Best Paper Honorable Mention**) ([link](#))
- [5] **E. C. Chau**, L. H. Lin, and N. A. Smith. 2020. Parsing with Multilingual BERT, a Small Corpus, and a Small Treebank. In *Findings of ACL: EMNLP*. ([link](#))

Education

University of Washington

M.S., Computer Science

June 2021 | Seattle, WA | GPA: 4.00

Thesis: *Specializing Multilingual Language Models: An Empirical Study*

B.S. with Distinction,

Computer Science (Data Science)

B.A., Linguistics

June 2020 | Seattle, WA | GPA: 3.98

Thesis: *Towards Resource-Efficient Contextual Word Representations for Parsing*

ETH Zürich

Exchange Student, Computer Science

Sept. 2019 – Feb. 2020 | Zürich, CH

Coursework

Natural Language Processing

NLP Capstone

Advanced Machine Learning

Artificial Intelligence

Reliable and Interpretable AI

Generative Models • Graphical Models

Information Retrieval

Quantum Computing

Data Structures • Algorithms

Operating Systems • Security

Skills

Technologies

Proficient

Bash, Git, Java, NumPy, Python, PyTorch,

SQL, Transformers

Familiar

C, C++, C#, Google Protobuf, Hack,

HTML/CSS, JavaScript, Mercurial, React,

Ruby, SCSS, Thrift

Languages

English (Native), Mandarin (Fluent),

Cantonese (Proficient), Spanish

(Intermediate), German (Beginner)

Service

Young Student Fellowship

High School Mentor

Sept. 2022 – Present